JavaScript Problem Solving

Day 1 Practice Set: 10 beginner problems with hints

Is PDF ka istemal kaise karein

Har problem ko in **7 steps** mein hal karo. Seedha code mat likho, pehle copy par logic banao:

1. Problem, Input, Output copy par apne alfaaz mein likho.

2. Ek chhota example khud kagaz par hal karo (insaan ki tarah).

3. Algorithm likho: numbered steps, Start se End tak.

4. Pseudocode likho: SET, FOR, IF, RETURN jaise words ke saath.

5. Dry run karo: table banao (i, current value, result).

6. Edge cases socho: khali input, ek element, negative numbers.

7. Ab JavaScript mein code likho aur test cases chalao.

Rules

- Pehle khud 20-30 minute koshish karo. Phir Hint 1, atko to Hint 2, phir Hint 3. Hints ek saath mat padhna.
- Jo built-in method (Math.max, reduce, includes ...) constraint mein mana hai, use mat karo. Maqsad logic seekhna hai.
- Function se hamesha return karo. console.log sirf test karne ke liye hai.
- Code ho jaye ya kahin atko to apna code Claude ko bhejo, wo hint ke saath check karega.
- Din mein 2-3 problems kaafi hain. Jaldi mat karo, har problem ka dry run zaroor karo.

Code bhejne se pehle checklist (aapki ab tak ki galtiyon se)

[ ] Loop: `i < arr.length` hai, <= to nahi? (warna aakhir mein undefined / NaN aata hai)

[ ] Comparison mein === hai, akela = to nahi? (`if (x = 0)` value rakhta hai, compare nahi karta)

[ ] `return` sahi jagah hai? Jawab sab elements dekhne ke baad chahiye to loop ke bahar, pehla match milte hi chahiye to loop ke andar.

[ ] Function return karta hai ya sirf print? Print karne se bahar wale console.log mein undefined aata hai.

[ ] Counter / total ki shuruaati value sahi hai? (sum ke liye 0, multiplication ke liye 1, khali array par kya hoga?)

Problem 1: Sum of 1 to N

Level: Easy

New concept: Accumulator pattern: ek total jo har loop mein barhta hai.

DESCRIPTION

Given a number **n**, calculate the sum of all whole numbers from 1 up to n using a for loop, and return the result.

INPUT / OUTPUT

Input: A non-negative integer n.
Output: One number: 1 + 2 + ... + n.

EXAMPLES

Input	Output	Note
`5`	`15`	1+2+3+4+5
`10`	`55`	
`1`	`1`	Only one number
`0`	`0`	Nothing to add
`100`	`5050`	

CONSTRAINTS

- Use a for loop.
- Do not use the formula n * (n + 1) / 2.
- Return the result, do not just print it.

STARTER CODE AND TEST CASES

```
function sumToN(n) {
  // your code here
}

console.log(sumToN(5));    // 15
console.log(sumToN(10));   // 55
console.log(sumToN(1));    // 1
console.log(sumToN(0));    // 0
console.log(sumToN(100));  // 5050
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye array ke sum wale problem jaisa hai, bas ab array nahi, 1 se n tak ke numbers hain.

Hint 2: Total ki shuruaati value kya hogi? Loop 1 se n tak chalao. Yahan <= sahi hai ya <? Soch kar likho.

Hint 3: n = 0 par loop ek baar bhi nahi chalta, to total ki shuruaati value hi jawab ban jati hai.

Problem 2: Multiplication Table

Level: Easy

New concept: Naya array banana aur push() se usme values daalna.

DESCRIPTION

Given a number **n**, return an array containing the **first 10 multiples** of n (n x 1, n x 2, ... n x 10).

INPUT / OUTPUT

Input: An integer n.

Output: An array of 10 numbers.

EXAMPLES

Input	Output	Note
`3`	`[3, 6, 9, 12, 15, 18, 21, 24, 27, 30]`	Table of 3
`7`	`[7, 14, 21, ..., 70]`	Table of 7
`1`	`[1, 2, 3, ..., 10]`	
`0`	`[0, 0, 0, ..., 0]`	Ten zeros
`-2`	`[-2, -4, ..., -20]`	Negatives work too

CONSTRAINTS

- Use a for loop.
- Do not use Array.from() or map().
- Return the array.

STARTER CODE AND TEST CASES

```
function tableOf(n) {
  // your code here
}

console.log(tableOf(3));   // [3, 6, 9, 12, 15, 18, 21, 24, 27, 30]
console.log(tableOf(1));   // [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
console.log(tableOf(0));   // [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle ek khali array banao: let result = []; Phir loop mein usme values daalo.

Hint 2: Loop i = 1 se 10 tak chalega. Har baar jo value array mein jani hai wo n aur i se banti hai. result.push(value) se daalo.

Hint 3: return loop ke bahar likhna, warna pehli value ke baad hi function khatam ho jayega.

Problem 3: Average of an Array

Level: Easy

New concept: Sum aur count ko jod kar naya result nikalna, aur divide by zero se bachna.

DESCRIPTION

Given an array of numbers, return the **average** (sum divided by how many numbers there are).

INPUT / OUTPUT

Input: An array of numbers, arr.

Output: One number: the average. Return 0 for an empty array.

EXAMPLES

Input	Output	Note
`[2, 4, 6]`	`4`	12 / 3
`[10]`	`10`	One element
`[1, 2]`	`1.5`	Decimals are fine
`[-2, 2]`	`0`	Negatives cancel out
`[]`	`0`	Empty array

CONSTRAINTS

- Use a for loop to find the sum.
- Do not use reduce().
- Do not divide by zero: handle the empty array first.

STARTER CODE AND TEST CASES

```
function average(arr) {
  // your code here
}

console.log(average([2, 4, 6]));   // 4
console.log(average([10]));        // 10
console.log(average([1, 2]));      // 1.5
console.log(average([-2, 2]));     // 0
console.log(average([]));          // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Average = total / kitne numbers hain. Total ka code aap sum wale problem mein likh chuke ho.

Hint 2: Kitne numbers hain, ye arr.length se pata chalta hai. Divide loop ke baad karna, andar nahi.

Hint 3: Khali array par 0 / 0 karoge to NaN aata hai. Function ke shuru mein ek if lagao jo is case ko pehle hi sambhal le.

Problem 4: Sum of Even Numbers

Level: Easy

New concept: Condition ke andar accumulator: sirf kuch numbers jodna.

DESCRIPTION

Given an array of integers, return the **sum of only the even numbers**. Ignore odd numbers.

INPUT / OUTPUT

Input: An array of integers, arr.

Output: One number: the sum of the even values.

EXAMPLES

Input	Output	Note
`[1, 2, 3, 4, 5, 6]`	`12`	2 + 4 + 6
`[1, 3, 5]`	`0`	No even numbers
`[2, 4]`	`6`	All even
`[-2, 3, 4]`	`2`	-2 + 4
`[]`	`0`	Empty array

CONSTRAINTS

- Use a for loop.
- Return the sum, do not just print it.

STARTER CODE AND TEST CASES

```
function sumEven(arr) {
  // your code here
}

console.log(sumEven([1, 2, 3, 4, 5, 6]));  // 12
console.log(sumEven([1, 3, 5]));           // 0
console.log(sumEven([2, 4]));              // 6
console.log(sumEven([-2, 3, 4]));          // 2
console.log(sumEven([]));                  // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye do purane problems ka mix hai: even count wala (condition) aur sum wala (total).

Hint 2: Even ki pehchan arr[i] % 2 === 0 hai. Is condition ke andar total mein kya jodna hai?

Hint 3: Count mein 1 jodte the, yahan arr[i] jodna hai. Aur return loop ke bahar hona chahiye.

Problem 5: Count Positive, Negative and Zero

Level: Easy

New concept: Teen counters aur if / else if / else, aur ek object return karna.

DESCRIPTION

Given an array of numbers, count how many are **positive**, how many are **negative** and how many are **zero**. Return the three counts in an object.

INPUT / OUTPUT

Input: An array of numbers, arr.
Output: An object: { positive: ?, negative: ?, zero: ? }

EXAMPLES

Input	Output	Note
`[3, -1, 0, 5, -7, 0]`	`{ positive: 2, negative: 2, zero: 2 }`	
`[1, 2, 3]`	`{ positive: 3, negative: 0, zero: 0 }`	All positive
`[-5]`	`{ positive: 0, negative: 1, zero: 0 }`	One element
`[0, 0]`	`{ positive: 0, negative: 0, zero: 2 }`	
`[]`	`{ positive: 0, negative: 0, zero: 0 }`	Empty array

CONSTRAINTS

- Use a for loop.
- Use if / else if / else (each number belongs to exactly one group).
- Return an object with exactly these three keys.

STARTER CODE AND TEST CASES

```
function countSigns(arr) {
  // your code here
}

console.log(countSigns([3, -1, 0, 5, -7, 0]));  // { positive: 2, negative: 2, zero: 2 }
console.log(countSigns([1, 2, 3]));             // { positive: 3, negative: 0, zero: 0 }
console.log(countSigns([]));                    // { positive: 0, negative: 0, zero: 0 }
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Teen alag counters chahiye: positive, negative, zero. Teeno ki shuruaati value kya hogi?

Hint 2: Har number teen mein se sirf ek group mein jata hai, isliye if (> 0), else if (< 0), else (baqi 0) use karo.

Hint 3: Aakhir mein return { positive: p, negative: n, zero: z }; (apne variable ke naam ke hisab se).

Problem 6: Largest of Three Numbers

Level: Easy

New concept: Sirf conditions se faisla karna (loop ke bina).

DESCRIPTION

Given **three numbers** a, b and c, return the largest one.

INPUT / OUTPUT

Input: Three numbers: a, b, c.

Output: One number: the largest of the three.

EXAMPLES

Input	Output	Note
`3, 9, 5`	`9`	b is largest
`10, 2, 7`	`10`	a is largest
`-1, -5, -3`	`-1`	Negatives
`4, 4, 4`	`4`	All equal
`1, 8, 8`	`8`	Two are tied

CONSTRAINTS

- Do not use Math.max().
- Do not put the numbers in an array.
- Return the number, do not just print it.

STARTER CODE AND TEST CASES

```
function maxOfThree(a, b, c) {
  // your code here
}

console.log(maxOfThree(3, 9, 5));     // 9
console.log(maxOfThree(10, 2, 7));    // 10
console.log(maxOfThree(-1, -5, -3));  // -1
console.log(maxOfThree(4, 4, 4));     // 4
console.log(maxOfThree(1, 8, 8));     // 8
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Sabse bade wale array problem ka idea yahan bhi chalta hai: pehle ek number ko largest maan lo.

Hint 2: largest = a rakho. Phir poocho: kya b > largest? Phir: kya c > largest? Har sawal ke baad largest update ho sakta hai.

Hint 3: Do alag if likho (else if nahi), taake b aur c dono check hon.

Problem 7: Count a Character in a String

Level: Easy

New concept: String ke har character ko ek target character se compare karna.

DESCRIPTION

Given a string and a single character, return **how many times** that character appears in the string. The match is case-sensitive.

INPUT / OUTPUT

Input: A string str and a one-character string ch.
Output: One number: how many times ch appears in str.

EXAMPLES

Input	Output	Note
`"banana", "a"`	`3`	
`"hello", "l"`	`2`	
`"hello", "z"`	`0`	Not present
`"Pakistan", "a"`	`2`	Capital P is a different character
`"", "a"`	`0`	Empty string

CONSTRAINTS

- Use a for loop.
- Do not use split(), match() or filter().
- Return the count, do not just print it.

STARTER CODE AND TEST CASES

```
function countChar(str, ch) {
  // your code here
}

console.log(countChar("banana", "a"));    // 3
console.log(countChar("hello", "l"));     // 2
console.log(countChar("hello", "z"));     // 0
console.log(countChar("Pakistan", "a"));  // 2
console.log(countChar("", "a"));          // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye vowels wale problem ka asaan version hai: wahan 5 letters se compare karna tha, yahan sirf ek (ch) se.

Hint 2: str[i] === ch se har character ko check karo. Counter kab barhega?

Hint 3: Comparison mein === use karo, aur counter ko loop ke andar barhao, return loop ke bahar.

Problem 8: FizzBuzz

Level: Easy+

New concept: Condition ka order, aur array mein numbers aur strings mix karna.

DESCRIPTION

Given a number **n**, return an array of the numbers from 1 to n, but: replace multiples of 3 with **"Fizz"**, multiples of 5 with **"Buzz"**, and multiples of both 3 and 5 with **"FizzBuzz"**.

INPUT / OUTPUT

Input: A non-negative integer n.
Output: An array of length n.

EXAMPLES

Input	Output	Note
`5`	`[1, 2, "Fizz", 4, "Buzz"]`	
`15`	`[1, 2, "Fizz", ..., 14, "FizzBuzz"]`	15 is a multiple of both
`3`	`[1, 2, "Fizz"]`	
`1`	`[1]`	Just one number
`0`	`[]`	Nothing to build

CONSTRAINTS

- Use a for loop.
- Return the array, do not just print it.

STARTER CODE AND TEST CASES

```
function fizzBuzz(n) {
  // your code here
}

console.log(fizzBuzz(5));   // [1, 2, "Fizz", 4, "Buzz"]
console.log(fizzBuzz(3));   // [1, 2, "Fizz"]
console.log(fizzBuzz(1));   // [1]
console.log(fizzBuzz(0));   // []
console.log(fizzBuzz(15));  // last item should be "FizzBuzz"
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Multiple ki pehchan % operator se hoti hai: i % 3 === 0 matlab i, 3 ka multiple hai.

Hint 2: 15 ke liye dono shartein sach hoti hain. Agar pehle sirf 3 wali condition check karoge to 15 par kya hoga? Condition ka order sochho.

Hint 3: Pehle 'dono ka multiple' (3 aur 5 dono) check karo, phir sirf 3, phir sirf 5, aur baqi mein number khud push karo.

Problem 9: Find the Index of a Number

Level: Easy+

New concept: Loop se mile hue index ko return karna, aur 'nahi mila' ke liye -1.

DESCRIPTION

Given an array and a target number, return the **index** of the **first** place the target appears. If the target is not in the array, return **-1**.

INPUT / OUTPUT

Input: An array arr and a number target.
Output: A number: the index, or -1.

EXAMPLES

Input	Output	Note
`[4, 8, 15, 16], 15`	`2`	15 is at index 2
`[4, 8, 15, 16], 4`	`0`	First element
`[4, 8, 15, 16], 23`	`-1`	Not found
`[5, 5, 5], 5`	`0`	First occurrence only
`[], 1`	`-1`	Empty array

CONSTRAINTS

- Use a for loop.
- Do not use indexOf() or findIndex().
- Use === for comparison.

STARTER CODE AND TEST CASES

```
function findIndexOf(arr, target) {
  // your code here
}

console.log(findIndexOf([4, 8, 15, 16], 15));  // 2
console.log(findIndexOf([4, 8, 15, 16], 4));   // 0
console.log(findIndexOf([4, 8, 15, 16], 23));  // -1
console.log(findIndexOf([5, 5, 5], 5));        // 0
console.log(findIndexOf([], 1));               // -1
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye 'number exists' wale problem jaisa hai. Wahan true return karte the, yahan kuch aur return karna hai.

Hint 2: Jaise hi arr[i] === target mile, loop ka i hi aapka jawab hai. Wahin return kar do.

Hint 3: return -1 loop ke bahar likho (jab poora array dekh liya aur nahi mila). Andar else mein mat likhna, warna pehle element par hi -1 return ho jayega.

Problem 10: Sum of Digits

Level: Medium-Easy

New concept: while loop, % 10 se aakhri digit nikalna, aur Math.floor() se digit hatana.

DESCRIPTION

Given a non-negative integer, return the **sum of its digits**.

INPUT / OUTPUT

Input: A non-negative integer n.

Output: One number: the sum of the digits of n.

EXAMPLES

Input	Output	Note
123	6	1 + 2 + 3
9	9	Single digit
0	0	
4059	18	4 + 0 + 5 + 9
1000	1	1 + 0 + 0 + 0

CONSTRAINTS

- Do not convert the number to a string.
- Use a loop (while is a natural choice here).
- Return the sum, do not just print it.

STARTER CODE AND TEST CASES

```
function sumDigits(n) {
  // your code here
}

console.log(sumDigits(123));   // 6
console.log(sumDigits(9));     // 9
console.log(sumDigits(0));     // 0
console.log(sumDigits(4059));  // 18
console.log(sumDigits(1000));  // 1
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Kisi bhi number ka aakhri digit n % 10 se milta hai. Jaise 123 % 10 = 3.

Hint 2: Aakhri digit hatane ke liye n ko 10 se divide karke decimal hata do: Math.floor(n / 10). Jaise 123 se 12 ban jata hai.

Hint 3: Ye do kaam tab tak dohrao jab tak n > 0 hai (while loop). Har baar aakhri digit total mein jodte jao. Pehle copy par 123 ka dry run karo.

Progress Tracker

Har problem ke saamne jo step mukammal ho, wahan tick (X) lagao.

#	Problem	Algorithm	Pseudocode	Dry run	Code	All tests pass
1	Sum of 1 to N					
2	Multiplication Table					
3	Average of an Array					
4	Sum of Even Numbers					
5	Count Positive, Negative and Zero					
6	Largest of Three Numbers					
7	Count a Character in a String					
8	FizzBuzz					
9	Find the Index of a Number					
10	Sum of Digits					

Bonus sawal (har problem ke baad khud se poocho)

- Is code ki time complexity kya hai? Kitni baar loop chalta hai?
- Is code ki space complexity kya hai? Kya naya array ya string bana?
- Best case aur worst case mein kya farq hai?
- Kya main isi problem ko doosre tareeqe se bhi hal kar sakta hoon?